

- 7) A technician is launching fireworks near the end of a show. Of the remaining fourteen fireworks, nine are blue and five are red. If she launches six of them in a random order, what is the probability that exactly four of them are blue ones?
- 8) A jar contains ten black buttons and six brown buttons. If nine buttons are picked at random, what is the probability that exactly five of them are black?
- 9) You are dealt five cards from a standard and shuffled deck of playing cards. Note that a standard deck has 52 cards and four of those are kings. What is the probability that you'll have at most three kings in your hand?
- 10) A bag contains six real diamonds and five fake diamonds. If six diamonds are picked from the bag at random, what is the probability that at most four of them are real?